

Problem Set 1

Total Points: 170 points

Deadline: 9:00 pm, September 22nd, 2026

All answers must be **handwritten**, then scanned or photographed, and submitted via Canvas.

Late submissions will incur a penalty of 5 points per day. Submissions more than 3 days late will not be accepted.

Problems from *Classical Mechanics* of John R. Taylor [12 × 10 points + 20 points]

Chapter 1: 1.32, 1.33, 1.39, 1.40, 1.47, 1.48, 1.49

Chapter 4: 4.2, 4.3, 4.22, 4.24, 4.34

Chapter 4: 4.38 (Write a python script to make the plot)

Problems from *Introduction to Classical Mechanics: With Problems and Solutions* of David Morin [2 × 10 points]

Page 77, 3.36 (Note: the plane is not movable)

Page 80, 3.53

Potential energies for positive power-law forces [10 points]

A particle moves in one dimension subject to the power-law force $F = -kx^n$, where the coefficient k is positive, and n is a positive integer. Please find the potential energy of the particle and also the maximum distance $x_{\max}$ it can reach from the origin, in terms of its maximum speed $v_{\max}$. The maximum distance is the **turning point** of the particle, because as the particle approaches this position it slows down, stops at $x_{\max}$, and turns around and heads in the opposite direction.
